



SmartHotel AI

AI-Powered Operating System for
Small Hotels & Restaurants



Built with LangGraph Multi-Agent Architecture

Confidential Business Plan — September 2026

Version 1.0 | Prepared for Internal Review

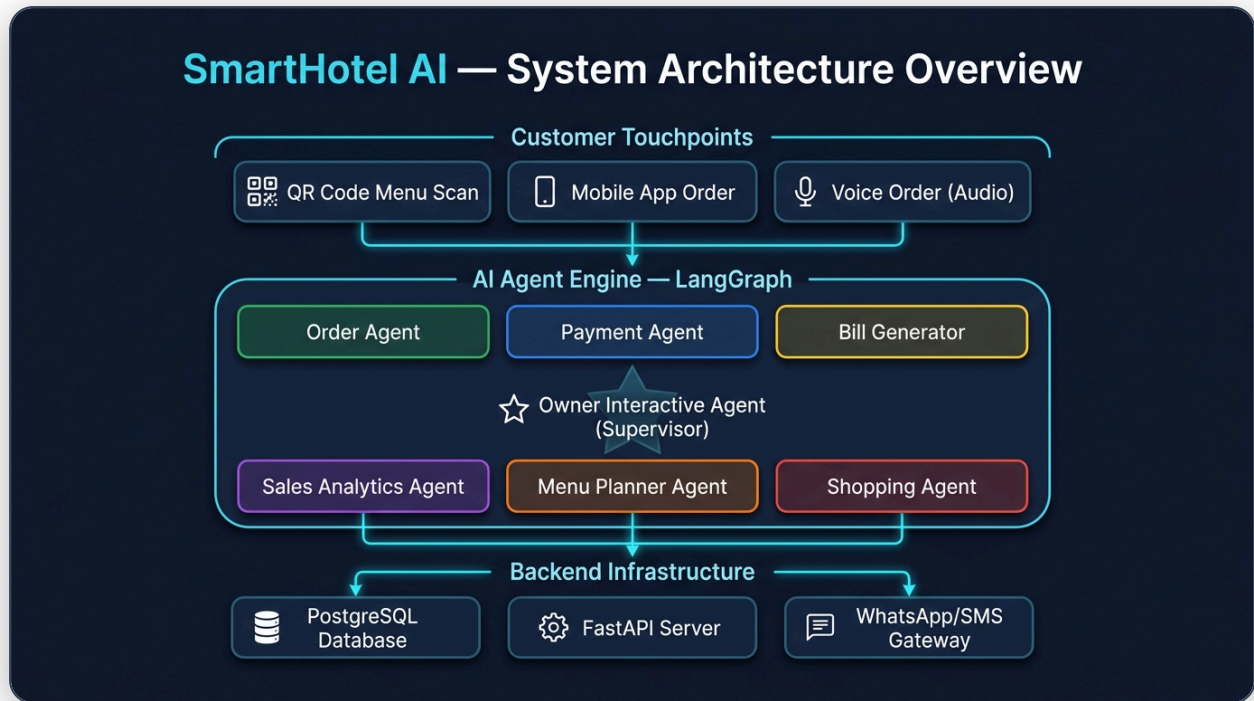


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System Architecture Overview

SmartHotel AI is built as a three-layer system connecting customer touchpoints to an AI agent engine powered by LangGraph, backed by robust infrastructure.



Architecture Layers

LAYER	COMPONENTS	PURPOSE
Customer Touchpoints	QR Code Scan, Mobile App, Voice Input	How customers interact with the system
AI Agent Engine	6 Specialized Agents + Supervisor	Intelligent processing & automation
Backend Infrastructure	PostgreSQL, FastAPI, WhatsApp Gateway	Data storage, API, notifications

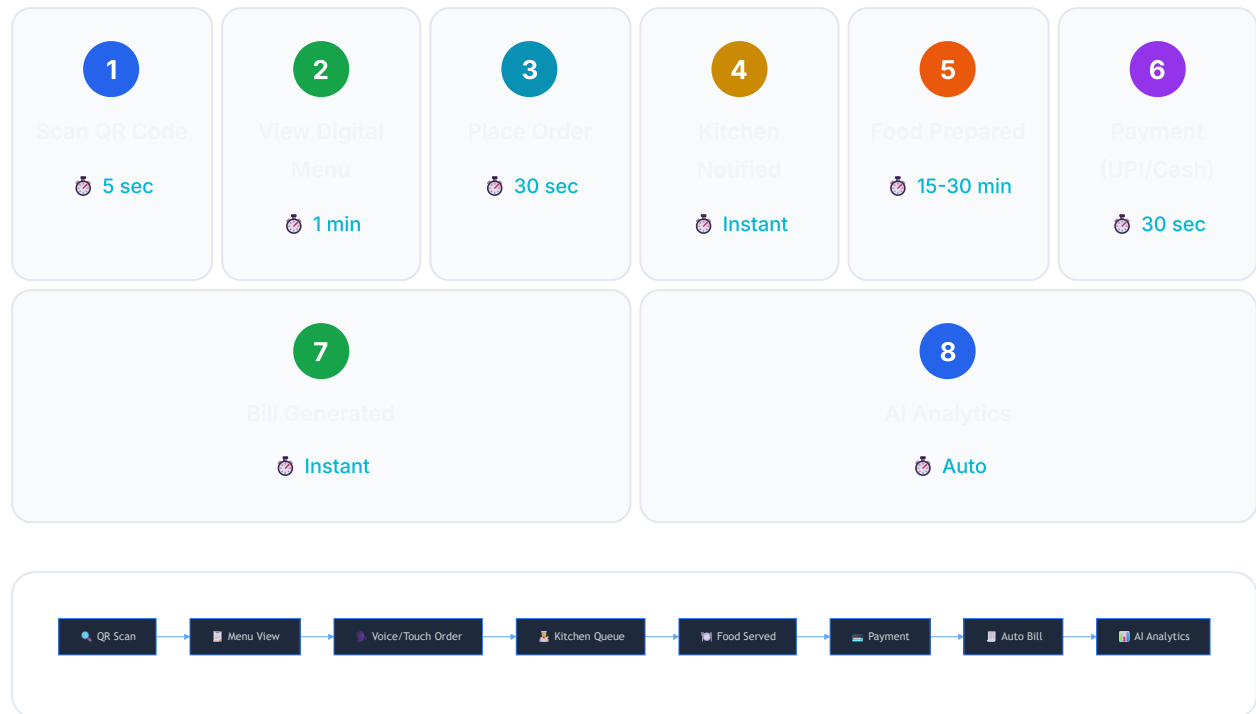
💡 **Key Design Principle:** Every hotel shares the same agent code. Personalization comes from hotel-specific data injected at runtime — no per-hotel model training

needed.

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Customer Order Journey

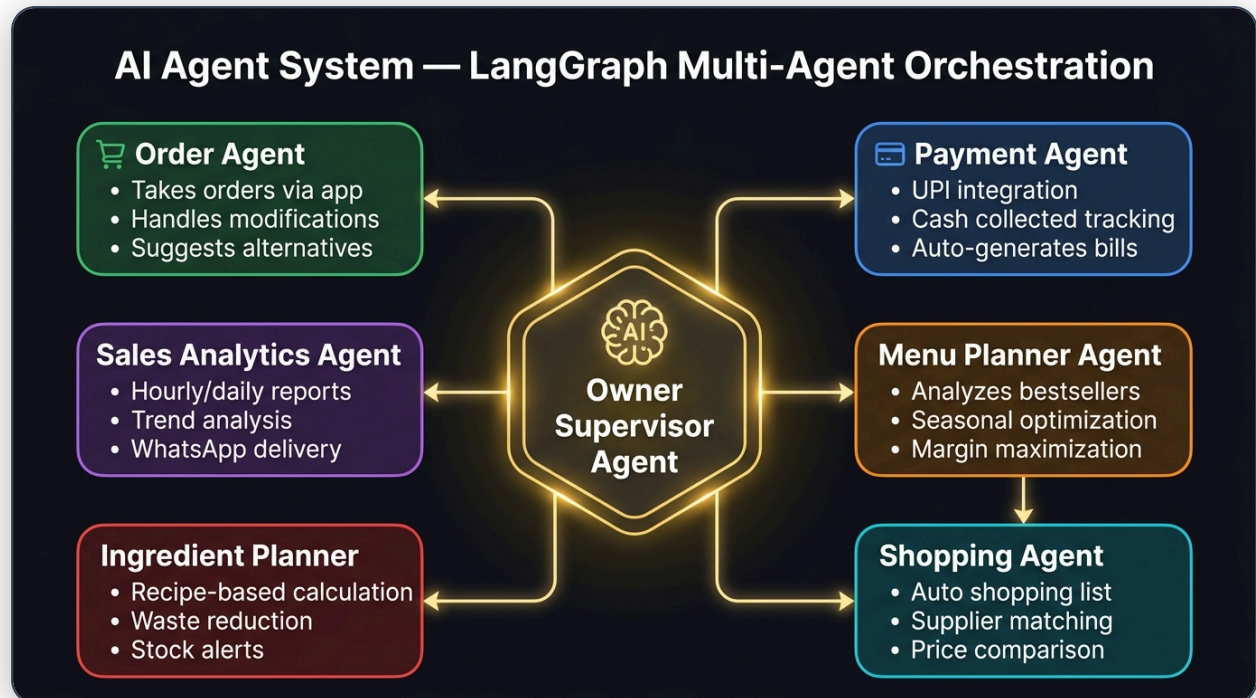
The complete order flow from the moment a customer sits down to when the AI captures analytics — fully automated, under 2 minutes to order.



🌟 **Customer Experience:** Total time from scan to order placed is under 2 minutes. Everything after that is fully automated by AI agents.

AI Agent System — LangGraph

Six specialized AI agents orchestrated by a central Supervisor agent, built on LangGraph's multi-agent framework for reliable, stateful processing.



Agent Details



Order Agent

- ▶ Takes orders via app (voice or touch)
- ▶ Handles order modifications
- ▶ Suggests alternatives for unavailable items
- ▶ Runs on every order (~100/day)



Payment Agent

- ▶ UPI integration via Razorpay
- ▶ Cash collected button & tracking
- ▶ Auto-generates digital bills
- ▶ Deterministic flow — no AI hallucination risk



Sales Analytics Agent

- ▶ Generates reports every 3 hours + daily summary



Menu Planner Agent

- ▶ Analyzes 30-day sales history
- ▶ Optimizes for seasonal ingredients

- ▶ Trend analysis & comparisons
- ▶ Delivered via WhatsApp/Push notifications
- ▶ Highlights top sellers & slow-moving items

- ▶ Maximizes profit margins
- ▶ Runs daily evening for next-day planning

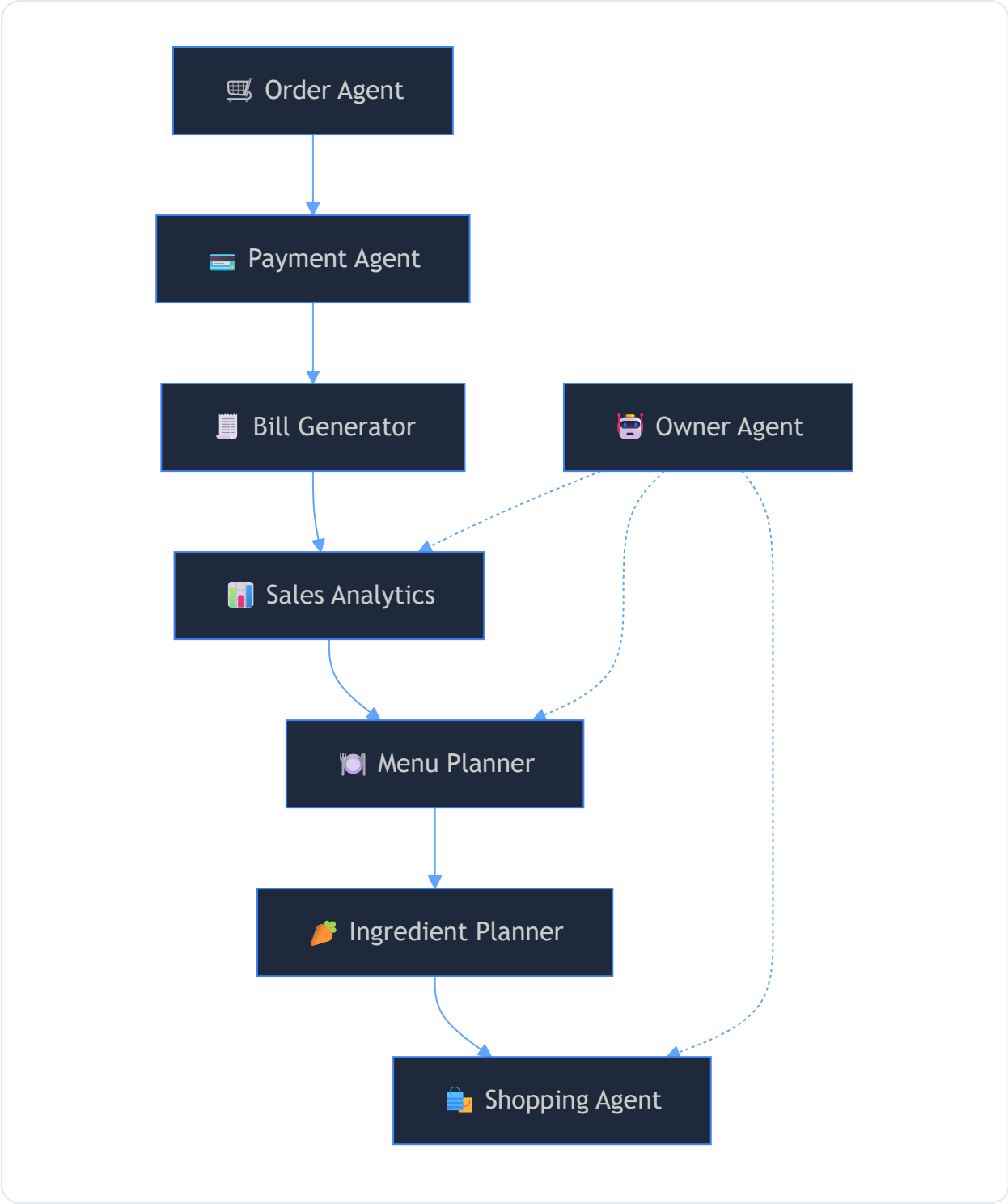
Ingredient Planner

- ▶ Recipe-based quantity calculation
- ▶ Waste reduction through precise planning
- ▶ Low-stock alerts
- ▶ Triggered after menu plan finalized

Shopping Agent

- ▶ Auto-generates shopping list
- ▶ Matches items to preferred suppliers
- ▶ Sends list via WhatsApp to suppliers
- ▶ Tracks purchase price history

Agent Chain — How They Work Together



Agent Trigger Schedule

AGENT	TRIGGER	FREQUENCY	PLAN
Order Agent	Customer places order	Every order (~100/day)	Free+
Payment Agent	Order completed	Every order	Free+

AGENT	TRIGGER	FREQUENCY	PLAN
 Bill Generator	Payment confirmed	Every order	Free+
 Sales Analytics	Cron job	Every 3 hrs + daily	Growth+
 Menu Planner	Scheduled / On-demand	Daily evening	Pro
 Ingredient Planner	Menu plan finalized	After planning	Pro
 Shopping Agent	Ingredient list ready	After ingredients	Pro
 Owner Interactive	Owner opens chat	On-demand	Growth+

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Revenue Model & Pricing Strategy

Three-Tier Pricing

FREE

₹0/month

Forever free

✓ Digital Menu (QR Code)

✓ 20 Orders/Day Limit

✓ Basic Bill Generation

🔗 Hook — Drive Adoption

★ MOST POPULAR GROWTH

₹999/month

Just ₹33/day — less than a chai

✓ Unlimited Orders

✓ Daily Sales Reports

✓ UPI + Cash Payments

✓ Owner Dashboard

✓ Basic Owner Chat Agent

🔗 Bread & Butter — 80% Revenue

PRO

₹2,499/month

₹83/day — full AI power

✓ All Growth Features

✓ AI Menu Planner

✓ AI Ingredient Planner

✓ AI Shopping Agent

✓ Voice/Audio Support

✓ Real-time Analytics

🔗 Profit Multiplier — High Margin

Why ₹999/Month is the Sweet Spot

FACTOR	REASONING
Daily cost	₹33/day — less than a customer's chai
Value delivered	Saves ₹1,000+/day in waste, time, and errors
ROI for owner	30x return — very easy decision

FACTOR	REASONING
Competitor price	Petpooja ₹1,000+ (no AI), POSist ₹2,000+
Our cost	₹300-600/hotel → 40-70% margin

Revenue Projection (18 Months)

STAGE	HOTELS	AVG REV/HOTEL	MONTHLY REVENUE	MONTHLY COST	PROFIT
Launch	10	₹999	₹9,990	₹8,000	₹1,990
Traction	50	₹1,200	₹60,000	₹25,000	₹35,000
Growth	200	₹1,400	₹2,80,000	₹70,000	₹2,10,000
Scale	500	₹1,500	₹7,50,000	₹1,50,000	₹6,00,000
Mature	1,000	₹1,600	₹16,00,000	₹2,50,000	₹13,50,000

💰 **At 500 hotels:** ₹6 Lakhs/month profit | **At 1,000 hotels:** ₹13.5 Lakhs/month profit

India's Food Service Industry — Key Numbers

75L+

Restaurants in
India

~90%

Have ZERO
Software

₹5.9L

Cr

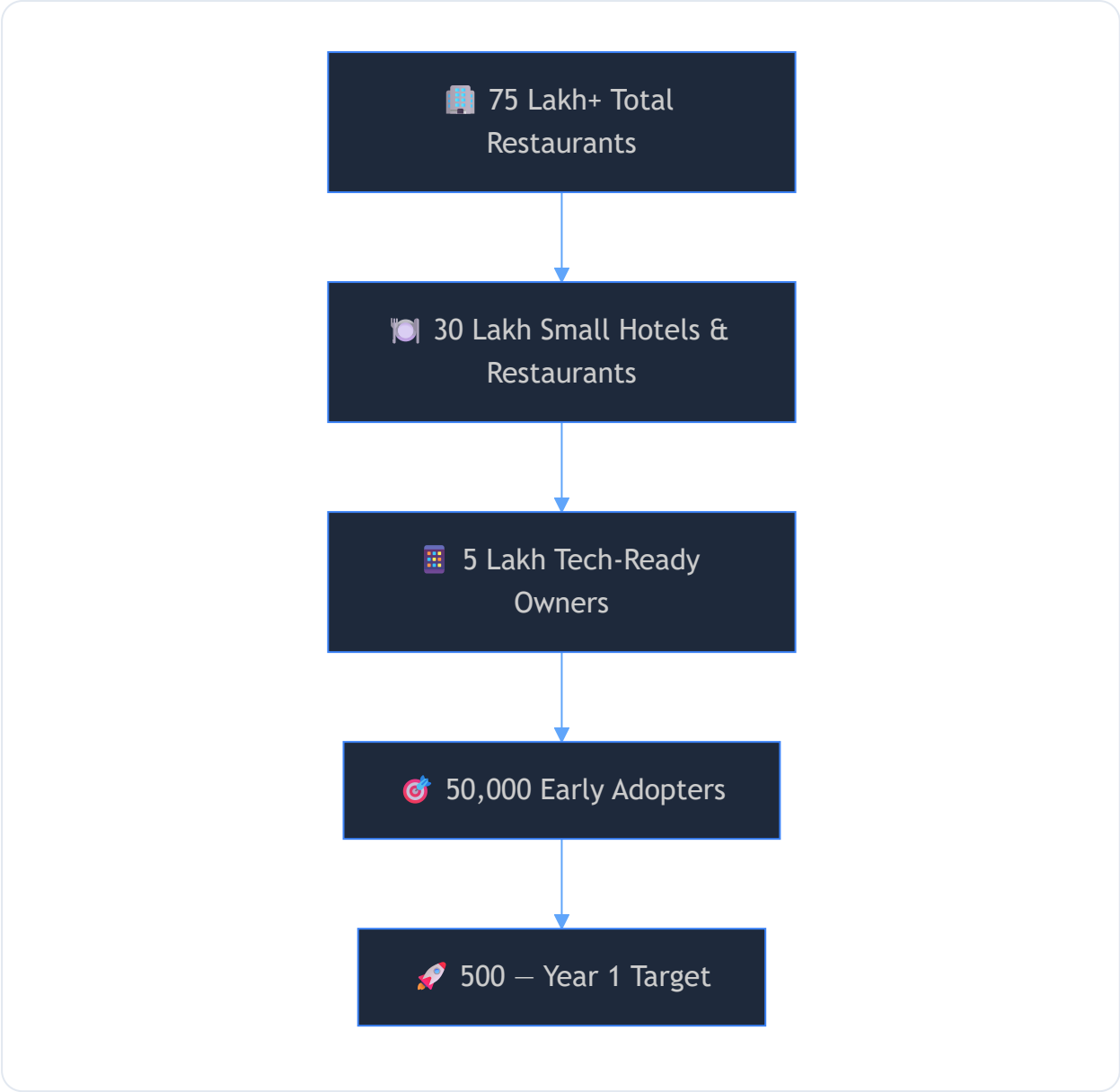
Industry Size
(2025)

22%

YoY

Digital Adoption
Growth

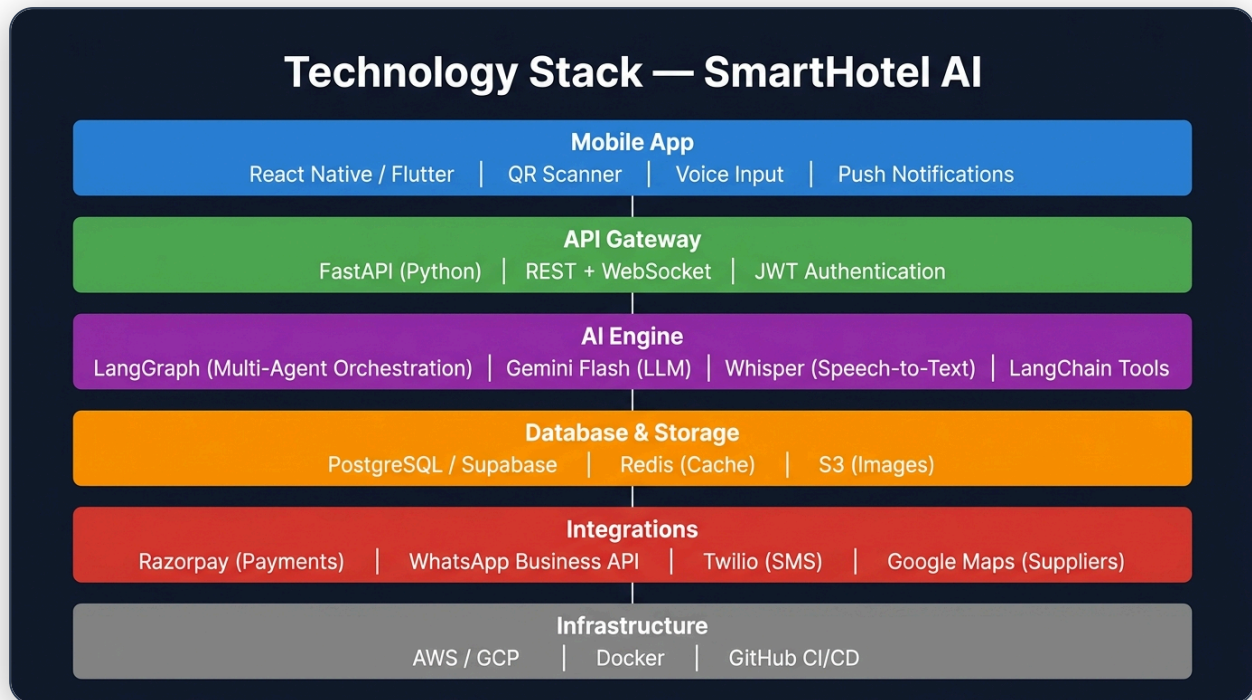
Target Market Funnel



Competitive Landscape

COMPETITOR	PRICE	AI FEATURES	OUR ADVANTAGE
Petpooja	₹1,000+/mo	❌ None	AI-native at similar price
POSist	₹2,000+/mo	❌ None	50% cheaper with more features
DotPe	Commission-based	❌ None	No commission, flat fee

COMPETITOR	PRICE	AI FEATURES	OUR ADVANTAGE
Zomato/Swiggy	20-30% commission	✗ None	In-house ops, not delivery
SmartHotel AI	₹999/mo	✓ 6 AI Agents	AI-native + Affordable + Voice-first



Stack Breakdown

LAYER	TECHNOLOGY	PURPOSE
 Mobile App	React Native / Flutter	Cross-platform iOS + Android
 API Gateway	FastAPI (Python)	REST + WebSocket, JWT auth
 AI Engine	LangGraph + Gemini Flash	Multi-agent orchestration
 Speech	Whisper (OpenAI)	Voice-to-text for orders
 Database	PostgreSQL / Supabase	Multi-tenant data storage
 Cache	Redis	Session & menu caching
 Payments	Razorpay	UPI + Card payments
 Messaging	WhatsApp Business API	Reports to owner

LAYER	TECHNOLOGY	PURPOSE
● Infra	AWS / GCP + Docker	Scalable cloud deployment

Why LangGraph? Your hotel system is inherently a graph — Order → Payment → Bill is a directed flow. LangGraph gives you deterministic control for payments (no hallucination risk) and AI flexibility for menu planning and analytics.

Hotel Onboarding — Live in 45 Minutes

 5 minutes

Step 1: Hotel Registration

Owner enters hotel name, cuisine type (Veg/Non-veg/Both), location, operating hours, and phone number via the mobile app.

 15 minutes

Step 2: Menu Setup

Enter menu items manually OR **scan existing paper menu with AI OCR**. AI auto-categorizes items into Starters, Main Course, Drinks, etc. Owner reviews and confirms.

 10 minutes

Step 3: Recipe Mapping

AI suggests standard recipes with ingredient quantities. Owner adjusts to match their actual recipes. "Chicken Biryani → Rice 200g, Chicken 250g, Onion 100g..."

 5 minutes

Step 4: Supplier Setup

Add regular suppliers with contact info and items they supply. "Raju Chicken Shop — Chicken, Mutton, Fish — 98xxxxx"

 5 minutes

Step 5: Go Live!

QR codes auto-generated for each table. The hotel is live — first order can be placed immediately!

AI-Assisted Onboarding Advantage

STEP	WITHOUT AI	WITH AI ASSIST	TIME SAVED
Menu entry (30 items)	45 minutes	10 minutes (OCR scan)	78%
Recipe mapping	60 minutes	10 minutes (AI suggestions)	83%
Supplier setup	15 minutes	5 minutes (structured form)	67%

Product Roadmap — 18 Months

● PHASE 1: MVP LAUNCH

Months 1–3 (Sep – Nov 2026)

- ◆ Digital Menu + QR Code generation
- ◆ Order Taking App (touch-based)
- ◆ Bill & Payment (UPI + Cash)
- ◆ Basic Sales Report (daily summary)

🎯 Target: 10 Pilot Hotels (Free)

● PHASE 2: AI LAYER

Months 4–6 (Dec 2026 – Feb 2027)

- ◆ Sales Analytics Agent (3-hourly reports)
- ◆ Menu Planner Agent
- ◆ Owner Interactive Chat Agent
- ◆ WhatsApp Integration

🎯 Target: 50 Paying Hotels

● PHASE 3: FULL SUITE

Months 7–12 (Mar – Aug 2027)

- ◆ Ingredient Planner Agent
- ◆ Shopping Agent
- ◆ Voice/Audio Support (Whisper)
- ◆ Multi-language Support

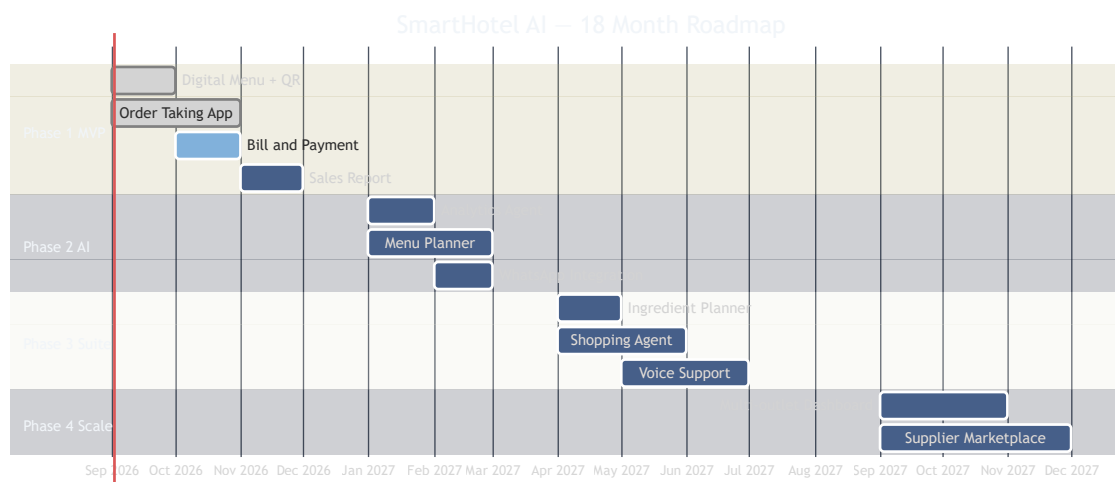
🎯 Target: 200–500 Hotels

● PHASE 4: SCALE

Months 13–18 (Sep 2027 – Feb 2028)

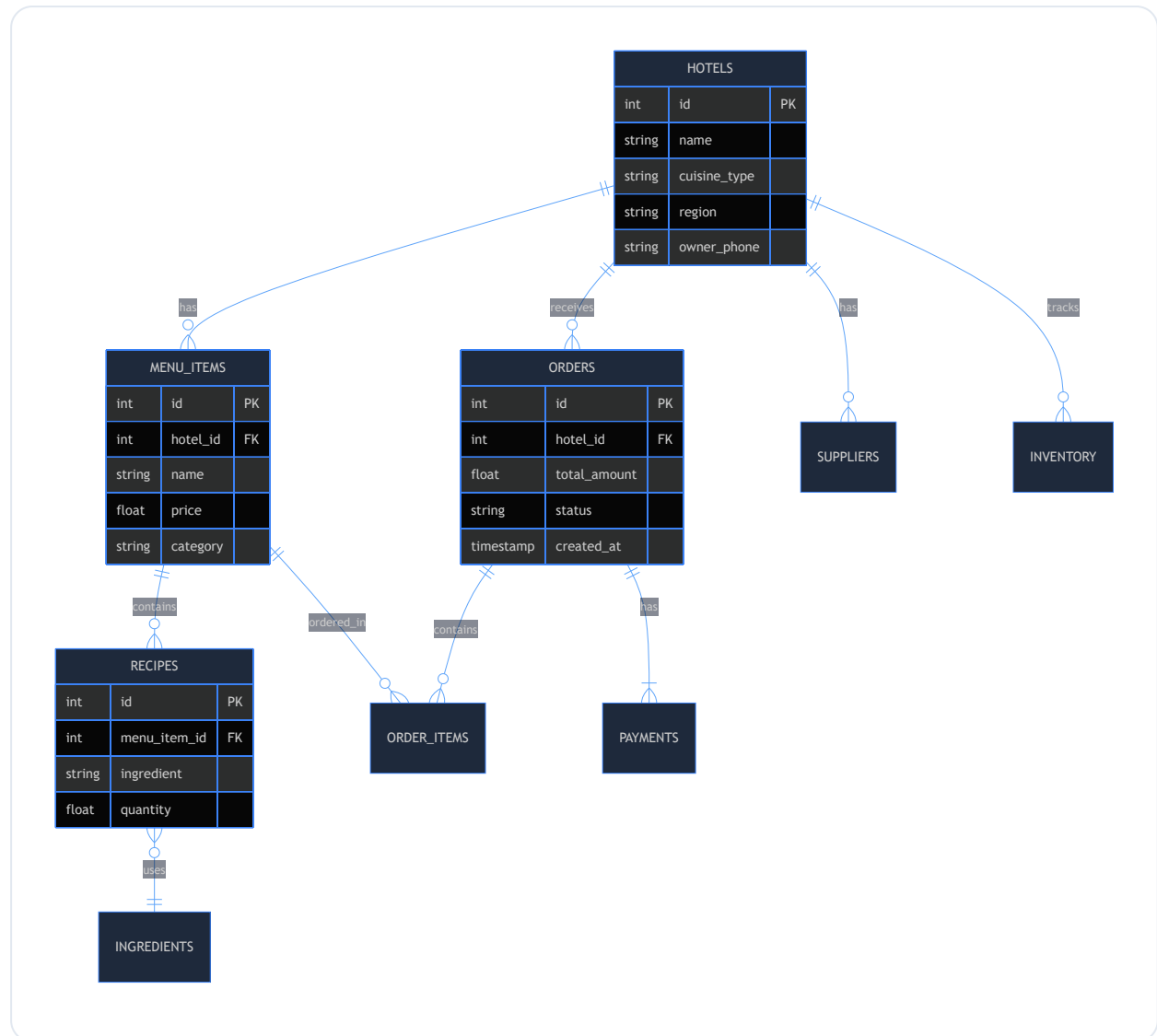
- ◆ Multi-outlet Owner Dashboard
- ◆ Supplier Marketplace
- ◆ Customer Loyalty Program
- ◆ Third-party POS API integration

🎯 Target: 1,000+ Hotels



Multi-Tenant Data Architecture

Single database serving all hotels, isolated by `hotel_id`. Maximum cost efficiency with complete data separation.



Key Tables

TABLE	PURPOSE	KEY COLUMNS
hotels	Hotel profile & config	name, cuisine, region, language, owner_phone

TABLE	PURPOSE	KEY COLUMNS
menu_items	Each hotel's menu	hotel_id, name, price, category, is_available
recipes	Ingredient mapping per dish	menu_item_id, ingredient, quantity, unit
orders	All orders placed	hotel_id, items, total, payment_mode, timestamp
suppliers	Hotel's regular suppliers	hotel_id, name, phone, items_supplied
inventory	Current stock levels	hotel_id, ingredient, current_stock, unit
ai_suggestions	Track AI recommendations	hotel_id, agent_type, suggestion, accepted
owner_preferences	Custom business rules	hotel_id, key, value

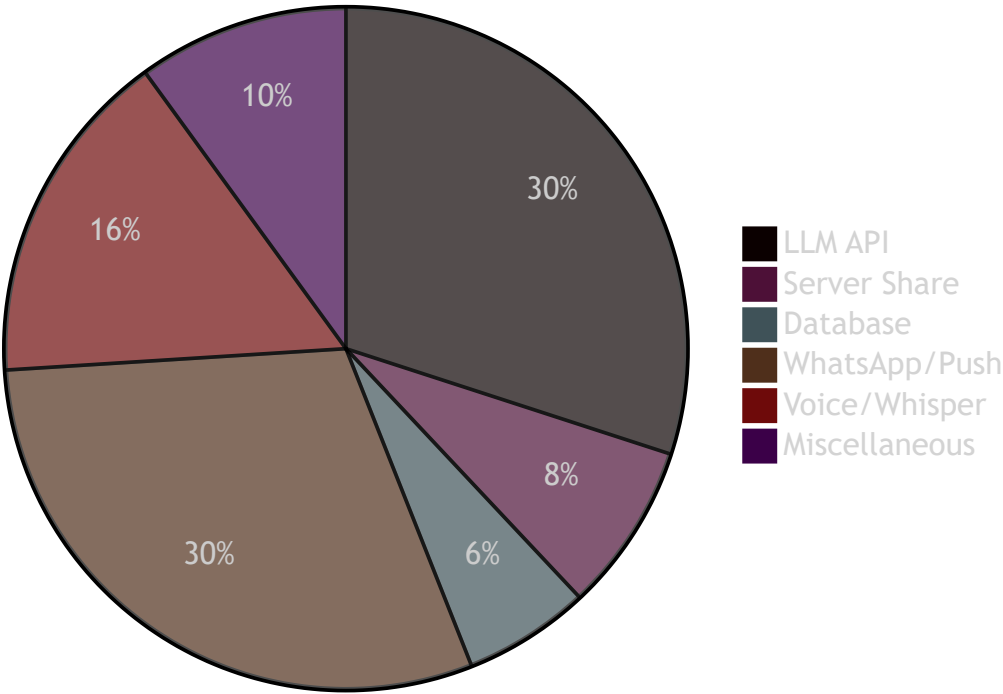
 **Data Isolation:** Every query filters by hotel_id. Hotel A can never see Hotel B's data. Single database, completely isolated data.

Cost Structure Per Hotel

Monthly Infrastructure Cost Breakdown

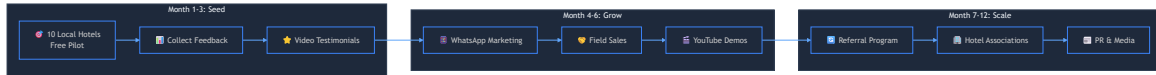
COMPONENT	COST/HOTEL/MONTH	NOTES
 LLM API (Gemini Flash)	₹80–200	~300 calls/day, cheapest tier
 Server (shared VPS)	₹20–50	1 server handles 100+ hotels
 Database (Supabase)	₹20–40	Shared multi-tenant
 Push Notifications	₹0–50	Firebase Cloud Messaging (free)
 WhatsApp (daily summaries only)	₹100–200	~30 msgs/month (not hourly)
 Voice/Whisper (Pro only)	₹0–200	Only Pro plan users
Total (Optimized)	₹300–600	Per hotel per month

Monthly Cost Distribution Per Hotel



Margin Analysis

PLAN	REVENUE	COST	MARGIN	MARGIN %
Free	₹0	₹100	-₹100	Loss leader
Growth (₹999)	₹999	₹400	₹599	60%
Pro (₹2,499)	₹2,499	₹600	₹1,899	76%



Channel Strategy

PHASE	CHANNEL	ACTION	TARGET
Seed	Direct outreach	Visit 20 local hotels, offer free pilot	10 pilots
Seed	Personal network	Friends & family who know hotel owners	5 warm leads
Grow	WhatsApp groups	Hotel owner communities, demo videos	200 impressions
Grow	YouTube	Regional language product demos	1,000 views
Grow	Field sales	Hotel-to-hotel visits with tablet demo	50 signups
Scale	Referral	₹500 off per successful referral	100 referrals
Scale	Hotel associations	Bulk partnership deals	200+ signups

The 30-Second Sales Pitch

"Sir, aapka restaurant ka digital menu, order system, payment, aur daily sales report — sab ek app mein. AI automatically bataega ki kal kya banana chahiye, kitna saman kharidna chahiye, aur kaunsa item sabse zyada bik raha hai. Sirf ₹33 per din. Ek chai se bhi sasta."

"Your restaurant's digital menu, order system, payment, and daily sales report — all in one app. AI will automatically tell you what to cook tomorrow, how much to buy, and which item sells the most. Just ₹33 per day. Cheaper than a cup of tea."

Risk Analysis & Mitigation

RISK	SEVERITY	IMPACT	MITIGATION STRATEGY
Low adoption — Owners resist technology	● HIGH	No revenue	Free tier + field demos + voice-first + regional language support
LLM hallucination — Wrong bill/payment amount	● HIGH	Trust breakdown	Payment flows are deterministic (no AI). Only analytics/planning use AI.
WhatsApp API cost — Eats into margins	● MEDIUM	Lower margins	Push notifications for routine updates, WhatsApp only for daily summaries
Competition — Petpooja/POSist add AI	● MEDIUM	Feature parity	Move fast, build brand loyalty, voice-first differentiator
Internet issues — Rural hotels, poor connectivity	● MEDIUM	App unusable	Offline-first app with background sync when online
Scale issues — Server overload at peak hours	● LOW	Slow performance	Auto-scaling cloud infra, queue-based agent processing, Redis caching

Executive Summary

Why SmartHotel AI Will Work

Right Market

75L+ restaurants in India, 90% with zero software — massive untapped opportunity

Right Price

₹33/day — cheaper than chai, delivers 30x ROI to the owner

Right Technology

LangGraph multi-agent system — deterministic where it matters, AI where it helps

Right Timing

AI costs dropping rapidly, Digital India adoption growing 22% YoY, post-COVID QR acceptance

Right Differentiation

AI-native (not bolted on), voice-first for adoption, affordable for small hotels

Key Metrics to Track

METRIC	MONTH 3	MONTH 6	MONTH 12	MONTH 18
Hotels onboarded	10	50	300	1,000
Monthly Revenue	₹0 (free pilots)	₹60K	₹4.5L	₹16L
Monthly Profit	-₹20K	₹35K	₹3.5L	₹13.5L
Churn Rate	<10%	<8%	<5%	<3%
NPS Score	>40	>50	>60	>70

Next Steps

- 1 Finalize MVP scope** — Confirm Phase 1 feature set
- 2 Design LangGraph architecture** — Nodes, edges, state schema, tools
- 3 Build prototype** — Order → Payment → Bill flow
- 4 Pilot with 5 local hotels** — Validate all assumptions
- 5 Iterate & Launch** — Based on real feedback

SmartHotel AI — AI-Powered Operating System for Small Hotels & Restaurants

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